

Contents

1	Task 1.....	3
1.1	A detailed overview of their product range, covering both high and low conductivity probes.3	
1.2	Technical specifications for each product.....	4
1.3	Information on materials used in construction (wetted materials, sensor materials).	5
1.4	Key differentiators, including any proprietary technologies or unique selling points.	6
1.5	Their approach to ensuring high precision across broad measurement ranges.	7
1.6	Solutions provided for use in corrosive, abrasive, and other harsh environments.	8
4.2	A comparative analysis of material performance in both corrosive and abrasive conditions.....	10

1 Task 1

1.1 A detailed overview of their product range, covering both high and low conductivity probes.

Summary Table of Competitors and Conductivity Sensor Models

Manufacturer	Product Family	Model	Sensor Type	Measurement Range	Applications
[REDACTED]	[REDACTED]	[REDACTED]	2-electrode, Inductive	0.01 $\mu\text{S/cm}$ to 2,000 mS/cm	Pure water, chemical processing, wastewater
			2-electrode	0.1 $\mu\text{S/cm}$ to 2000 mS/cm	Chemical processing, ultrapure water
			2-electrode, 4-electrode	1 $\mu\text{S/cm}$ to 300,000 $\mu\text{S/cm}$	Pharmaceutical, wastewater, food-grade water
			2-electrode, Inductive	0.001 $\mu\text{S/cm}$ to 2,000 mS/cm	Wastewater, chemical, pure water
			Inductive	1 $\mu\text{S/cm}$ to 2,000 mS/cm	Hygienic environments, CIP/SIP
			4-electrode	0.01 $\mu\text{S/cm}$ to 500 mS/cm	Food, beverage, chemical, pharmaceutical

This summary includes detailed comparisons of 2-electrode, 4-electrode, and inductive sensor types, organized by major manufacturers and models, along with their corresponding measurement ranges and typical applications.

1.2 Technical specifications for each product.

This table summarizes the key technical specifications, materials, and applications of major conductivity probes by manufacturer, covering 2-electrode, 4-electrode, and inductive sensor types.

Manufacturer	Product Model	Sensor Type	Conductivity Range	Temperature Range	Pressure Rating	Materials	Key Features	Applications	Source
		2-Electrode	0.01 µS/cm to 2000 µS/cm	-20°C to 150°C	Up to 12 bars		Fast response with integrated temperature sensor; corrosion resistant	Pure water, ultrapure water, pharmaceutical applications	
		Inductive	100 µS/cm to 2000 mS/cm	-20°C to 130°C	Up to 12 bars	Stainless steel,	High resistance to aggressive chemicals; no fouling due to electrodeless design	Chemical processing, wastewater treatment	
		2-Electrode	0.1 µS/cm to 2000 µS/cm	-20°C to 180°C	Up to 30 bars		High precision for low-conductivity measurements; advanced diagnostics	Ultrapure water, chemical processing, power generation	
		2-Electrode	0.1 µS/cm to 500 mS/cm	-20°C to 200°C	Up to 30 bars		FDA-compliant; rapid temperature compensation	Pharmaceutical water, ultrapure water, food and beverage	
		4-Electrode	1 µS/cm to 300,000 µS/cm	-20°C to 150°C	Up to 16 bars		High chemical compatibility; wide range of measurement	Wastewater treatment, general water monitoring	
		2-Electrode	0.001 µS/cm to 500 µS/cm	-20°C to 100°C	Up to 10 bars	steel,	Ultra-low conductivity measurement; resistant to corrosion in pure water environments	Laboratory, ultrapure water	
		Inductive	0.1 µS/cm to 2,000 mS/cm	-20°C to 150°C	Up to 20 bars		High resistance to corrosive media; inductive sensor prevents fouling	Chemical processing, wastewater, aggressive environments	
		Inductive	1 µS/cm to 2000 mS/cm	-20°C to 130°C	Up to 25 bars		High chemical resistance; suitable for in hygienic applications	Food and beverage, pharmaceutical	
		4-Electrode	0.1 µS/cm to 500 mS/cm	-20°C to 120°C	Up to 10 bars		Designed for sanitary applications with integral Tri-Clamp fitting for easy installation	Food and beverage, chemical processing	

1.3 Information on materials used in construction (wetted materials, sensor materials).

Manufacturer	Product Model	Sensor Type	Wetted Materials	Sealing Material	Applications	Source
		2-Electrode		(FDA-approved)	Pure water, ultrapure water, chemical processing	
		Inductive		N/A	Chemical processing, wastewater treatment	
		2-Electrode		N/A	Ultrapure water, chemical processing	
		2-Electrode		N/A	Pharmaceutical water, ultrapure water, food and beverage	
		4-Electrode		N/A	Wastewater treatment, general water monitoring	
		2-Electrode		N/A	Laboratory, ultrapure water	
		Inductive		N/A	Chemical processing, wastewater	
		Inductive		N/A	Food and beverage, pharmaceutical	

1.4 Key differentiators, including any proprietary technologies or unique selling points.

Manufacturer	Product Model	Sensor Type	Key Differentiators	Proprietary Technologies
		2-Electrode	Wide measuring range for multiple applications; IP68-rated connector system ensures reliability	technology for broad use in ultrapure to saline solutions; VarioPin connector system
		Inductive	Highly resistant to fouling and corrosion; designed for harsh chemical environments	Electrodeless design prevents electrode degradation in aggressive media
		2-Electrode	Advanced diagnostics and self-monitoring for process reliability	Self-monitoring diagnostics for real-time sensor health tracking
		2-Electrode	FDA-compliant materials; rapid temperature compensation for accurate real-time measurements	enables digital integration with process control systems
		4-Electrode	Resistant to fouling; wide measurement range with high accuracy	for seamless integration with digital systems and wide conductivity range
		2-Electrode	Ultra-low conductivity measurement; corrosion-resistant materials for pure water applications	design provides high precision in ultrapure environments
		Inductive	High resistance to aggressive chemicals and fouling; suitable for corrosive environments	Inductive design prevents wear and minimizes maintenance through self-cleaning capabilities
		Inductive	Hygienic design for food and pharma industries; non-contact measurement for long life in aggressive conditions	Inductive technology prevents direct contact and extends sensor lifespan in CIP/SIP applications

1.5 Their approach to ensuring high precision across broad measurement ranges.

Manufacturer	Product Model	Sensor Type	Approach to Ensuring High Precision	Source
		2-Electrode	The sensor uses [REDACTED] technology to provide precise measurements from ultrapure water to higher conductivity ranges. [REDACTED] ensures stable signal transmission, reducing errors caused by [REDACTED]	
		Inductive	The inductive design minimizes [REDACTED] providing reliable measurements in harsh chemical environments. It is ideal for aggressive applications that require stable, long-term accuracy.	
		2-Electrode	The advanced diagnostics system offers real-time monitoring and correction for sensor health, ensuring consistent precision over time. The [REDACTED] ensure durability and stable performance in fluctuating environments.	
		2-Electrode	Uses [REDACTED] technology to integrate digital data transmission with process control systems, enabling continuous real-time monitoring. The sensor also features automatic temperature compensation for accurate measurements in fluctuating conditions.	
		4-Electrode	The 4-electrode design compensates for polarization, improving accuracy across a wide measurement range. Additionally, Arc technology ensures seamless digital integration for real-time accuracy.	
		2-Electrode	The [REDACTED] electrode [REDACTED] and small cell constant provide ultra-precise measurements in pure and ultrapure water, especially in laboratory settings.	
		Inductive	The inductive design eliminates	

			polarization errors and prevents fouling, providing stable, long-term measurements in high-conductivity and chemically aggressive environments.	
		Inductive	The non-contact inductive technology avoids electrode fouling and ensures long-lasting accuracy, even in demanding CIP/SIP processes. It uses advanced digital integration for precise real-time monitoring.	

1.6 Solutions provided for use in corrosive, abrasive, and other harsh environments.

Manufacturer	Product Model	Sensor Type	Solutions for Corrosive, Abrasive, and Harsh Environments	Source
			Designed with [REDACTED] materials, which provide corrosion resistance in chemically aggressive environments like ultrapure and deionized water. The inductive sensor design eliminates electrode contact, making it ideal for use in corrosive chemical environments such as wastewater and acid-based processes. [REDACTED] provide excellent durability and corrosion resistance in aggressive chemical and abrasive environments. [REDACTED] and [REDACTED] resistance to chemicals and cleaning agents, making the sensor ideal for food, beverage, and pharmaceutical environments. [REDACTED] offer high corrosion resistance in wastewater and industrial environments. [REDACTED] provide excellent	

International		Electrode	corrosion resistance for pure and ultrapure water applications in lab settings.	
			and provide strong resistance to aggressive chemicals, making this inductive sensor ideal for corrosive and abrasive environments in chemical processing. Hygienic design with and materials offers high chemical resistance, suitable for harsh cleaning environments like CIP/SIP processes.	

4.2 A comparative analysis of material performance in both corrosive and abrasive conditions.

Manufacturer	Product Model	Materials Used	Performance in Corrosive Conditions	Performance in Abrasive Conditions	References
[REDACTED]	[REDACTED]	[REDACTED]	Excellent chemical resistance, [REDACTED] construction. Ideal for handling aggressive chemicals in pharmaceutical and water applications	Moderate abrasion resistance; [REDACTED] offers some protection against wear but is not optimized for highly abrasive conditions	[REDACTED]
		[REDACTED]	High corrosion resistance, particularly in environments with strong acids, alkalis, and wastewater	High abrasion resistance, [REDACTED] is known for its robustness in industrial settings, offering durability against aggressive and abrasive particles	[REDACTED]
		[REDACTED]	Superior chemical resistance, with [REDACTED] long-term durability in highly corrosive chemical processes	Moderate resistance to abrasion, [REDACTED] protection and provides limited resistance to mechanical wear	[REDACTED]
		[REDACTED]	High resistance to chemical corrosion, especially in pharmaceutical and food processing environments that use caustic	Moderate abrasion resistance, [REDACTED] provides durability in environments with mild mechanical	[REDACTED]

			cleaning agents	wear	
			Exceptional chemical resistance, making it suitable for aggressive media like wastewater and industrial chemicals	High abrasion resistance, Hastelloy is renowned for its resilience in high-wear environments (	
			Effective corrosion resistance for less aggressive environments, such as ultrapure water; not suitable for strong acids.	Low abrasion resistance, is fragile and prone to damage from abrasive particles	
			High resistance to corrosive chemicals, such as acids and oxidizing agents, making it suitable for industrial chemical processing	High resistance to abrasion, both and provide durability against mechanical wear and abrasive particles	
		PTFE	Excellent chemical resistance, particularly suited for cleaning processes in food and pharma environments	High abrasion resistance, provides long-lasting durability in environments with aggressive particles and mechanical wear	